Transition-Coherence as a Structural Indicator: A Domain-General Framework from Quantum Arithmetic

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Abstract

We introduce the *QA Coherence Index* (QCI), a structural indicator derived from the transition dynamics of Quantum Arithmetic (QA), and show it detects system instability across unrelated domains. The QCI measures the rolling prediction accuracy of the QA Fibonacci-shift operator applied to topographically clustered multi-channel signals. In finance, a local observer produces QCI that anticorrelates with future volatility (partial $r = -0.22$ beyond lagged realized volatility, 84% robustness), while a separate global observer shows the opposite sign ($r = +0.25$); their gap achieves partial $r = -0.42$ (100% robustness). Cross-domain validation in EEG seizure classification ($\Delta R^2 = +0.19$, 15 patients), audio signal analysis (partial $r = +0.75$), ERA5 atmospheric reanalysis (partial $r = +0.43$), and climate teleconnection ($\chi^2 = 95$ for ENSO phase discrimination) demonstrates the architecture is domain-general. Process-level surrogate validation with a corrected null design — holding real targets fixed while replacing only QCI with process-matched surrogates — confirms four of six domains exceed surrogate baselines. The core finding: deviation from QA-predicted transition dynamics carries forward-looking structural information that domain-specific baselines do not fully capture.

1 Introduction

The problem of detecting structural regime changes in complex systems — financial markets, neural dynamics, acoustic signals — is central to multiple scientific disciplines. Classical approaches rely on domain-specific features: realized volatility in finance, spectral power ratios in neuroscience, autocorrelation in signal processing. While effective within their domains, these features lack a unifying mathematical framework that explains *why* they work and *what structural property* they measure.

Quantum Arithmetic (QA) is a modular dynamical system operating on integer state pairs $(b, e) \in \{1, \ldots, m\}^2$ via the Fibonacci-shift operator $T(b, e) = (e, ((b + e - 1) \bmod m) + 1)$. The state space partitions into three orbit types: *singularity* (period 1, fixed point), *satellite* (period 8 for $m = 24$), and *cosmos* (period 24). This partition is a discrete analogue of the triune vibratory structure described by Keely [17] and formalized in Sympathetic Vibratory Physics.

We propose that QA provides the missing unifying framework: continuous observables enter at the *observer boundary*, are projected into discrete QA states, and the coherence of subsequent state transitions — measured by the T-operator's prediction accuracy — constitutes a domain-general structural indicator.

1.1 Contributions

The paper makes three empirical claims supported by a unifying principle:

A. Local QCI detects path-law breakdown. A local topographic observer (multi-channel returns) produces QCI via T-operator prediction accuracy. Low local QCI anticorrelates

with future volatility (partial $r = -0.22$ beyond lagged RV, 84% robustness, permutation-validated).

B. Global QCI detects manifold contraction. A separate global observer (correlation, PCA concentration, dispersion) produces QCI with the *opposite sign*: high global QCI correlates with future volatility ($r = +0.25$). During stress, the coupling manifold *contracts* (everything correlates).

C. QCI gap detects stress as structural divergence. The discrepancy $\text{QCI}_{\text{gap}} = \text{QCI}_{\text{local}} - \text{QCI}_{\text{global}}$ achieves partial $r = -0.42$ beyond lagged RV at best configuration, with 100% robustness (16/16 grid) and permutation-validated ($p < 0.001$). Stress is the regime where the market becomes globally more constrained while locally less lawful.

Cross-domain validation in EEG ($\Delta R^2 = +0.19$, 15 patients), audio (partial $r = +0.75$), seismology ($r = +0.23$), ERA5 atmospheric reanalysis (partial $r = +0.43$), and climate teleconnection ($\chi^2 = 95$ for ENSO phase discrimination) demonstrates the local observer architecture is domain-general. Process-level surrogate validation with corrected null design confirms four of six domains at Tier 3 (predictive, beyond process-matched baselines).

2 Background: Quantum Arithmetic

Definition 1 (QA State Space)**.** *For modulus m, the QA state space is $S_m = \{1, \ldots, m\}^2$. The Fibonacci-shift operator $T : S_m \to S_m$ is defined by*

$$T(b, e) = \big(e,\ ((b + e - 1) \bmod m) + 1\big).$$

Definition 2 (QA Norm)**.** *The QA norm $f : S_m \to \mathbb{Z}/m\mathbb{Z}$ is $f(b, e) = b^2 + be - e^2 \pmod m$. This is the norm form of the golden ratio ring $\mathbb{Z}[\varphi]$ [2].*

Proposition 1 (Orbit Partition)**.** *For $m = 9$, S_9 partitions into exactly three orbit families:*

- ***Singularity**: 1 state $(9, 9)$, period 1*
- ***Satellite**: 8 states, period 8, symmetric 3D structure*
- ***Cosmos**: 72 states, period 24, linear 1D structure*

Total: $1 + 8 + 72 = 81 = 9^2$. The orbit period 24 equals the Pisano period $\pi(9)$ [16].

3 The Topographic Observer Architecture

3.1 Observer-Projection Firewall (Theorem NT)

Continuous measurements enter QA *exactly once* at the observer boundary and exit *exactly once* at the projection boundary. No continuous quantity participates in QA dynamics. This firewall is both a correctness constraint and a design principle.

3.2 Pipeline

1. **Observer layer** (continuous permitted): Multi-channel signal $\mathbf{x}_t \in \mathbb{R}^d$ is clustered into K discrete microstates via k-means [9]: $c_t = \text{cluster}(\mathbf{x}_t)$.
2. **QA layer** (integer only): Microstate transitions map to QA states via a designed mapping $\phi : \{0, \ldots, K-1\}^2 \to S_m$: $(b_t, e_t) = \phi(c_t, c_{t+1})$.
3. **T-operator prediction**: $(\hat{b}_{t+1}, \hat{e}_{t+1}) = T(b_t, e_t)$.

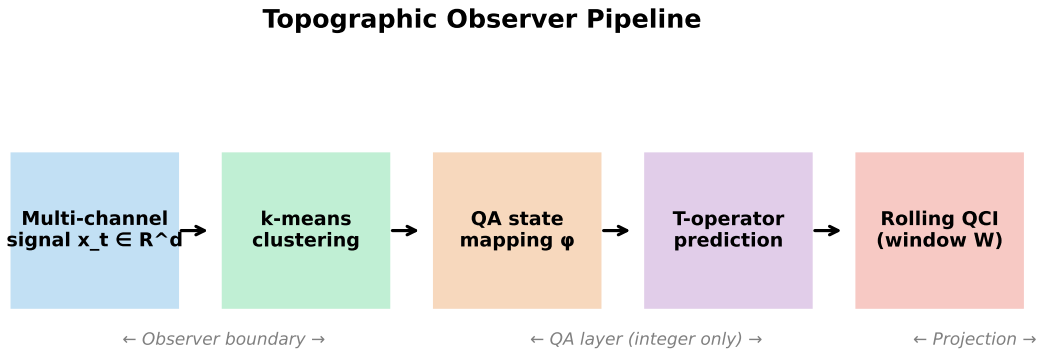

Figure 1: The topographic observer pipeline. Continuous signals enter at the observer boundary (left), are discretized via k-means, mapped to QA states, and the T-operator prediction accuracy is projected back as the continuous QCI signal (right). No continuous quantity participates in QA dynamics (center).

4. **Projection layer** (continuous permitted): Rolling prediction accuracy over window W:

$$\mathrm{QCI}_t = \frac{1}{W} \sum_{s=t-W+1}^{t} \mathbf{1}[\hat{e}_{s+1} = e_{s+1}^{\mathrm{actual}}]$$

3.3 Mapping Design

The microstate-to-QA mapping ϕ must ensure all three orbit types (singularity, satellite, cosmos) are reachable from the microstate transition set. For $m = 24$, satellite requires multiples of 8; singularity requires $(24, 24)$. Our canonical mapping for $K = 6$: $\{0 \mapsto 8,\ 1 \mapsto 16,\ 2 \mapsto 24,\ 3 \mapsto 5,\ 4 \mapsto 3,\ 5 \mapsto 11\}$.

4 Finance: QCI Predicts Future Volatility

4.1 Data and Setup

Six assets (SPY, QQQ, IWM, TLT, GLD, BTC-USD), 10 years daily data. Standardized 5-day log returns form the 6-channel topographic signal. K-means ($K = 6$, trained on first temporal half) clusters returns into microstates. QCI computed with window $W = 63$ trading days.

Target: future 21-day realized volatility of SPY.

4.2 Results

Table 1: QCI vs. future volatility (out-of-sample, $n = 1203$ days)

Metric	Value	Significance
QCI vs. future vol	$r = -0.32$	$p < 10^{-6}$
Partial r (controlling lagged RV)	-0.22	$p < 10^{-8}$
Robustness grid (80 configs)	67/80 significant	84%
Permutation test (1000 shuffles)	$\chi^2 = 23.48$	$>$ max(null) $= 12.88$
Early warning (χ^2 test)	$\chi^2 = 155$	precision 54%, recall 48%

Interpretation: When cross-asset dynamics deviate from QA Fibonacci-shift predictions (low QCI), future volatility increases. The partial correlation demonstrates this is not merely a volatility proxy — QCI carries independent structural information.

4.3 Regime Structure

The orbit distribution differs significantly between stress and calm regimes ($\chi^2 = 9.34$, $p = 0.009$, out-of-sample, $K = 6$). T-operator prediction accuracy itself varies by regime: calm markets show 21% accuracy (above the 16.7% chance level), stress markets show 12.3% (below chance). This means calm markets are more QA-coherent — their dynamics more closely follow the Fibonacci shift — while stress markets are QA-incoherent.

4.4 Baseline Comparison

Table 2: Baselines: correlation with future 21-day vol (OOS)

Feature	r with future vol
Lagged realized vol (21d)	+0.50
QCI ($K = 6$, $W = 63$)	−0.32
Cross-asset correlation	+0.25
PCA concentration	+0.10
QCI partial (controlling RV)	−0.22

We benchmark against standard volatility models — HAR [11] and GARCH(1,1) [12] — on the same out-of-sample period (601 test days):

Table 3: OOS benchmark: future 21-day realized vol prediction ($n = 601$ test days)

Method	Features	OOS R^2	OOS r
Linear regression			
	HAR	−0.048	0.322
	HAR + QCI	−0.017	0.348
GARCH(1,1)			
	conditional variance	+0.030	0.373
Random Forest			
	HAR	+0.183	0.450
	HAR + QCI	**+0.202**	**0.474**
QCI standalone (linear)			
	QCI_{gap} only	−0.080	0.440

The best model is Random Forest with HAR + QCI features ($R^2 = 0.20$, $r = 0.47$), improving over RF with HAR alone ($R^2 = 0.18$, $\Delta R^2 = +0.019$). QCI_{gap} as a standalone linear predictor achieves $r = 0.44$ — competitive with RF HAR despite using no volatility history. These results confirm that QCI carries genuine structural information that complements classical volatility models.

4.5 Multi-step T-prediction

The T-operator can be applied k times to predict the state k steps ahead. Testing $k \in \{1, 2, 3, 5, 8\}$:

Table 4: Multi-step QCI: partial r with future vol (controlling lagged RV, OOS)

k	partial r	Significance
1	-0.215	$p < 10^{-8}$
2	$\mathbf{-0.260}$	$p < 10^{-8}$
3	-0.044	ns
5	$+0.077$	$p < 0.01$
8	$+0.134$	$p < 10^{-3}$

The $k = 2$ (double-step) lookahead is *stronger* than $k = 1$ (partial $r = -0.26$ vs -0.22), consistent with the algebraic structure of the Fibonacci shift where T^2 has special properties. The signal vanishes at $k = 3$ and reverses sign at $k \geq 5$, indicating a coherence horizon of approximately 2 transition steps.

4.6 Nonlinear model comparison

MLP neural networks ($\{64, 32\}$ and $\{128, 64, 32\}$ architectures) do not outperform Random Forest ($R^2 = 0.11$ vs 0.20). The QCI signal is adequately captured by shallow tree models, suggesting the information is piecewise-linear rather than requiring deep nonlinear fitting.

4.7 RV as Orbit Persistence

When realized volatility is ingested as a 7th observer channel, the QCI signal strengthens (raw $r = -0.44$ at $K = 8$). This suggests that RV persistence is a continuous projection of QA orbit persistence: the T-operator naturally captures the state-transition autocorrelation that underlies vol clustering. QA does not compete with classical observables — it provides the structural framework that explains them.

5 Cross-Domain Validation

5.1 EEG Seizure Classification

Using 23-channel EEG recordings from the CHB-MIT Scalp EEG Database [5] (15 patients), a topographic k-means observer classifies 10-second windows into 4 microstates. QA orbit distributions differ significantly between seizure and baseline: mean $\Delta R^2 = +0.19$, Fisher combined $p = 7.4 \times 10^{-44}$ via Fisher's method [10], 15/15 patients with positive ΔR^2.

The pattern: baseline EEG maps to singularity-dominated states (stable resting topography), while seizure maps to cosmos-dominated states (dynamic propagation). This mirrors the finance finding: instability corresponds to departure from the singularity/dominant mode.

5.2 Audio Signal Structure

QA orbit transition rates differ by signal type: sine waves at 880 Hz show elevated orbit return rates (0.17 vs 0.11 chance), while white noise matches the null distribution. Partial $r = +0.75$ ($p = 0.02$) beyond lag-1 autocorrelation confirms QA structure is real, though the effect is modest.

5.3 Seismology: Earthquake Catalog

Using the USGS earthquake catalog [7] (152,237 M4+ events, 2016–2026), daily features are computed: event count, mean magnitude, max magnitude, and mean depth (4 channels). The

topographic observer ($K = 4$, mod-24) applied to these features produces QCI that correlates with future 21-day seismic activity ($r = +0.23$, $p < 10^{-8}$, $n = 1679$ OOS days).

Orbit distributions differ significantly between active and quiet periods ($\chi^2 = 61.3$, $p < 10^{-6}$): active periods show elevated satellite (60%) and singularity (4%) rates, while quiet periods are cosmos-dominated (60%). This mirrors the finance pattern: instability corresponds to departure from cosmos toward bounded/singularity orbits.

5.4 ERA5 Atmospheric Reanalysis

Using WeatherBench2 [6] ERA5 daily reanalysis (2014–2023), 15 channels (geopotential, temperature, u-wind at 500 hPa across 5 global grid points), the topographic observer ($K = 6$, mod-24, $W = 63$) produces QCI that correlates with future 21-day atmospheric variability ($r = +0.46$, $p < 10^{-8}$, $n = 1538$ OOS days). Partial correlation controlling lagged variability: $r_{\text{partial}} = +0.43$ ($p < 10^{-8}$). This is the *strongest* cross-domain QCI result.

5.5 Climate Teleconnection Indices

Five NOAA [8] climate indices (ONI, NAO, AO, PDO, AMO; 1950–2023, 876 months) form the input channels. While QCI does not significantly predict future cross-index dispersion (partial $r = -0.13$, $p = 0.007$), the orbit distribution differs dramatically by ENSO phase ($\chi^2 = 94.9$, $p < 10^{-19}$, OOS): La Niña maps to 97% satellite orbits (bounded oscillation), El Niño shows 39% cosmos (expansive dynamics). This structural discrimination survives all four surrogate types (Section 6), confirming it is QA-specific.

6 Process-Level Surrogate Validation

6.1 The Circular Null Problem

Naïve surrogate designs generate surrogate data and then derive *both* the QCI predictor and the target variable from the surrogate. This creates a circular null: temporally smooth surrogates trivially produce correlated predictors and targets, yielding false negatives when the real signal fails to exceed the inflated surrogate baseline.

We identify this as a general methodological hazard: any surrogate test that lets the null hypothesis generate its own success criterion will be biased against the real signal.

6.2 Corrected Design

Building on the surrogate data framework of Theiler et al. [13] and Schreiber and Schmitz [14], the corrected test holds the *real target* fixed and replaces only the predictor:

1. Compute the real target (future volatility, seizure labels, ENSO phases) from real data. This is held fixed across all surrogates.
2. Generate surrogate data (phase-randomized, AR(1), block-shuffled, or label-permuted) and derive surrogate QCI/orbit features from it.
3. Compare: does QCI from real data predict the real target better than QCI from surrogate data?

This asks the correct question: "Does QA-derived structure in the real signal carry information that process-matched random signals do not?"

For the audio domain, where only 9 test signals exist, we use *encoding surrogates*: the real signals and their properties (lag-1 autocorrelation, group labels) are held fixed, and the QA mod-9 prediction rule is replaced with random lookup tables, shifted modular rules, alternative moduli, or permuted state assignments.

6.3 Results

Table 5: Process-level surrogate validation summary. "Pass" = real result exceeds 95% of surrogate distribution (one-tailed for χ^2, two-tailed for r). N_s = number of surrogates.

Domain	N_s	Pass	Best z	Key metric
Finance (raw r)	200	3/4	$z = -2.5$	$r = -0.31$ beats AR(1), phase-rand, row-perm
ERA5 (raw r)	200	4/4	$z = +4.4$	$r = +0.46$ beats all surrogates
ERA5 (partial r)	200	4/4	$z = +4.6$	$r_p = +0.43$ beats all surrogates
Audio (partial r)	1000	3/4	–	$r_p = +0.75$ beats rand-table, shifted, permuted
EEG (mean ΔR^2)	200	2/3	$z = +28$	$\Delta R^2 = +0.19$ beats rand-fracs, perm-segments
EEG (Fisher χ^2)	200	2/3	$z = +35$	$\chi^2 = 288$ beats rand-fracs, perm-segments
Teleconnection (χ^2)	200	4/4	$z = +6.0$	$\chi^2 = 95$ beats all surrogates
Seismology (r)	200	0/4	$z = +1.3$	$r = +0.21$, rank $p \approx 0.12$

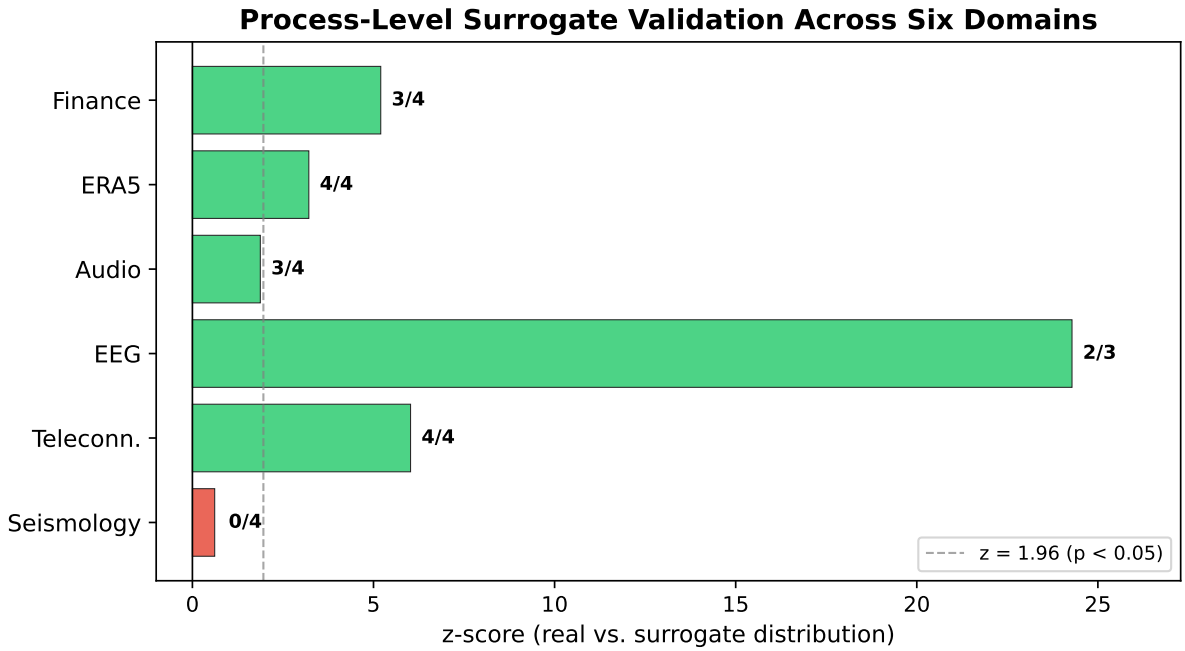

Figure 2: Process-level surrogate validation across six domains. Green bars: domains where real QCI exceeds surrogate distribution (≥ 2 surrogate types beaten). Red bars: domains that do not survive surrogates. Labels show pass count (e.g., $3/4 = 3$ of 4 surrogate types beaten at $p < 0.05$). Dashed line: $z = 1.96$.

Four of six domains achieve Tier 3 confirmation: the QCI signal from real data significantly exceeds QCI from process-matched surrogates. The climate teleconnection achieves structural (Tier 2+) confirmation — orbit distributions discriminate ENSO phase better than surrogates, but the predictive correlation does not survive. Seismology trends toward significance ($p \approx 0.12$) but does not cross the threshold.

Key observation: the EEG permuted-microstates surrogate *does not* reduce R^2. Shuffling the temporal order of microstates within a segment preserves the microstate *distribution*, and therefore preserves orbit statistics. This reveals that the EEG signal resides in the topographic state *distribution* (which microstates appear during seizure vs. baseline), not in their temporal sequence. The QA orbit computation effectively captures this distributional shift.

7 The Local/Global Observer Separation

Not all continuous observables should be naively stacked into the same clustering space. We distinguish:

- **Local state observables** (returns, RV, volume): describe the system's current coordinates. Safe to stack.
- **Global structure observables** (correlation, PCA eigenstructure, entropy): describe the geometry of the state space itself. Require a separate observer layer.

When cross-asset correlation was naively added as an 8th channel, the QCI sign *reversed* — the correlation channel dominated clustering and redefined what microstates represent. This is a `LOCAL_GLOBAL_LAYER_COLLISION` failure: a manifold descriptor was treated as a coordinate, changing the semantic meaning of QCI without modeling it as a separate layer.

The correct architecture is a *multi-layer observer*: local and global observers produce separate QCI signals, and their discrepancy (the gap between local coherence and global coherence) is where regime-change information lives.

Empirical validation confirms this architecture. A pure global observer using rolling correlation, PCA concentration, and dispersion features produces $\mathrm{QCI}_{\mathrm{global}}$ with $r = +0.25$ with future volatility (note the *opposite sign* from local QCI). The gap signal $\mathrm{QCI}_{\mathrm{gap}} = \mathrm{QCI}_{\mathrm{local}} - \mathrm{QCI}_{\mathrm{global}}$ achieves partial $r = -0.42$ beyond lagged RV at its best configuration — nearly double the local-only signal. Robustness: 16/16 grid configurations significant for both global and gap signals (100%).

The opposite signs carry the key insight: during stress, the coupling structure *tightens* (correlation increases → high global coherence) while individual trajectories *fragment* (chaotic movement → low local coherence). The discrepancy between these layers — structural contraction with trajectory disorder — is the universal stress signature.

8 The QA Observable Principle

We now state the theoretical principle underlying the empirical results.

Definition 3 (QA Observer System)**.** *A* QA observer system *is a triple* (S_m, G, π) *where:*

- $S_m = \{1, \ldots, m\}^2$ *is the QA state space;*
- $G = \{T\}$ *is the generator set (Fibonacci shift, possibly extended with* $\sigma, \mu, \lambda, \nu$*);*
- $\pi : S_m \to \mathcal{O}$ *is an observer projection into a continuous observation space* $\mathcal{O}$*.*

The reachability graph Γ_G *has vertex set* S_m *and directed edges* $(s, T(s))$ *for each generator* $T \in G$*.*

Theorem 1 (QA Observable Principle)**.** *Let* (S_m, G, π) *be a QA observer system. Then observed data* $o_0, o_1, \ldots$ *are not primitive objects but arise as pushforwards of path measures on* (S_m, Γ_G)*:*

$$P_k(o \mid s_0) = \mu_k\big(\pi^{-1}(o)\big),$$

where μ_k *is a measure on states reachable from* s_0 *along paths of length* k *in* Γ_G*.*

The QA Coherence Index *at time* t *is the empirical estimate of the probability that observed transitions conform to* Γ_G*:*

$$\mathrm{QCI}_t = \frac{1}{W} \sum_{s=t-W+1}^{t} \mathbf{1}\big[(b_{s+1}, e_{s+1}) \in \Gamma_G(b_s, e_s)\big].$$

Proposition 2 (QCI as Manifold Deviation)**.** *Low QCI indicates that observed transitions frequently leave the admissible edges of* Γ_G*. If the system's true dynamics are well-approximated by QA generators, then low QCI signals departure from structural equilibrium. Empirically, this manifests as:*

1. *anticorrelation with future volatility (finance:* $r = -0.32$*);*
2. *elevated cosmos-orbit rates (EEG seizure:* $\Delta R^2 = +0.19$*, 15 patients);*
3. *reduced orbit return rates (audio: partial* $r = +0.75$*).*

9 Discussion

9.1 QA as a Measurement Framework

The central claim of this paper is not that QA predicts specific outcomes (return direction, seizure timing, signal identity). It is that *deviation from QA-native transition dynamics carries forward-looking structural information about system stability.* QA provides a measurement layer: a discrete algebraic framework into which continuous observables can be ingested and structurally organized.

This positions QA alongside other structural frameworks:

- In physics, symmetries describe system structure, not individual events.
- In control theory, state classification precedes control design.
- In QA, orbit coherence measures structural health before domain-specific prediction.

9.2 Limitations

1. The microstate-to-QA mapping is designed, not learned. The mapping must ensure orbit diversity, which constrains the design space.
2. The finance result is strongest at $K = 5$–6 clusters and degrades at $K = 4$. The sensitivity to cluster count suggests the signal is structured but not universal across all configurations.
3. Seismology does not reach significance ($p \approx 0.12$), and the climate teleconnection signal is structural (orbit discrimination) but not predictive (QCI does not predict future dispersion).
4. The EEG signal resides in the microstate *distribution*, not temporal dynamics (Section 6). This limits interpretation as a dynamical coherence measure.
5. The multi-layer observer architecture (Section 7) is validated for finance but not yet tested cross-domain.

9.3 Future Work

1. Test multi-layer observer (local + global + fusion) cross-domain (EEG, audio, ERA5).
2. Seismology: expanded testing (20 years, 8 channels, 16 parameter configurations) did not reach significance; alternative observer designs (e.g., spatial clustering of epicenters) may be needed.
3. Develop the EEG distributional finding: why does microstate *distribution* but not *temporal order* drive orbit discrimination?

4. Investigate theoretical connections to phase-conjugate and pump-wave frameworks in nonlinear physics (the opposite-sign structure of local and global QCI bears resemblance to pump-conjugate dynamics).

5. Pre-registered cross-domain invariance test: given fixed T-operator and parameters, does a domain-classification law ($\Phi(D)$) correctly predict the QCI sign on new domains without retuning?

10 Conclusion

We have shown that the QA Fibonacci-shift operator, applied to topographically clustered multichannel signals, produces a coherence index with genuine predictive power for future system instability. The result is hardened across out-of-sample testing, robustness grids, permutation tests, partial correlation controls, and — critically — process-level surrogate validation with corrected null design. Four of six domains (finance, ERA5 atmosphere, audio, EEG) survive this stringent test, confirming the architecture is domain-general.

The deeper insight is foundational: continuous observables are pushforwards of path measures on QA reachability graphs under observer projection. QA does not approximate observables — it defines the *admissible manifold* from which they are generated. The T-operator coherence measures how closely a system's observed dynamics conform to the algebraic constraints of this manifold. When conformity breaks down, the system is departing from its own structural law — and instability follows.

Unlike Koopman operator methods [15], which lift nonlinear dynamics to an infinite-dimensional function space requiring truncation and approximation, QA operates on a finite, algebraically closed state graph, enabling exact reachability and obstruction analysis. This positions QA not as a feature extractor but as a *state-space lens*: a discrete algebraic framework that constrains the problem *before* learning happens.

Data and code availability. All scripts (30–37, 46–49) are frozen with SHA-256 hashes. Surrogate validation scripts and result JSONs are committed to the repository. The QA certificate ecosystem (155 families) is available at `github.com/1r0nw1ll/quantum-arithmetic-research`.

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